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30 Mar, 2026 Shift 1



Agniveervayu Intake 01 2027 30th March 2026

Date of Exam	30/03/2026
Exam Time	9:00 AM - 10:25 AM
Entry (Post Name)	Agniveervayu
Stream Applied	Both (Science Subjects and other than science subjects)

Section : Physics

Q.1 An adiabatic process occurs in a system where 200 J of work is done on the gas. What is the change in internal energy?

- Ans**
- ☒ 1. -200 J
 - ☒ 2. 0 J
 - ☒ 3. 200 J
 - ☒ 4. 400 J

Q.2 A point charge $2\text{ }\mu\text{C}$ produces an electric field E_1 at a distance r . Another point charge $8\text{ }\mu\text{C}$ produces an electric field E_2 at a distance $2r$. What is the ratio $E_2 : E_1$?

- Ans**
- ☒ 1. 1 : 4
 - ☒ 2. 1 : 2
 - ☒ 3. 2 : 1
 - ☒ 4. 1 : 1

Q.3 Which material property is obtained from the slope of the linear portion of the stress-strain curve?

- Ans**
- ☒ 1. Ultimate tensile strength
 - ☒ 2. Ductility
 - ☒ 3. Yield strength
 - ☒ 4. Young's modulus

Q.4 A scientist observes that when a light ray travels from medium A to medium B, the angle of refraction is larger than the angle of incidence. What can be concluded?

- Ans**
- ☒ 1. Medium B is optically rarer
 - ☒ 2. Both media A and B have equal refractive indices
 - ☒ 3. No refraction occurred
 - ☒ 4. Medium B is optically denser

Q.5 If the acceleration due to gravity on the Earth's surface is 9.8 m/s^2 , which statement is true for a 10 kg object on the Moon and the Earth?

- Ans**
- ☒ 1. Its weight on the Moon is 81.7 N higher than its weight on the Earth
 - ☒ 2. Its weight on the Earth is 16.3 N higher than its weight on the Moon
 - ☒ 3. Its weight on the Earth is 81.7 N higher than its weight on the Moon
 - ☒ 4. Its weight on the Moon is 16.3 N higher than its weight on the Earth

Q.6 A car accelerates uniformly from rest to 20 m/s in 10 seconds. How far does it travel during this time?

- Ans**
- ☒ 1. 50 m
 - ☒ 2. 150 m
 - ☒ 3. 100 m
 - ☒ 4. 200 m

Q.7 Let F be the force between two point charges, $+3 \mu\text{C}$ and $-3 \mu\text{C}$ placed 10 cm apart in vacuum. What will be the force if vacuum is replaced by dielectric medium of relative permittivity 2 ?

- Ans**
- ☒ 1. $F/4$
 - ☒ 2. $2F$
 - ☒ 3. $F/2$
 - ☒ 4. $4F$

Q.8 A physical constant has dimensions $[M L T^{-3}]$. Which of the following physical quantities could represent this constant?

- Ans**
- ☒ 1. Force per unit area
 - ☒ 2. Momentum per unit time
 - ☒ 3. Energy per unit volume
 - ☒ 4. Power per unit length

Q.9 When viewed from above, if the current in a circular loop flows clockwise, where does the magnetic dipole moment point?

- Ans**
- ☒ 1. Outward radially from the center
 - ☒ 2. Downward from the plane
 - ☒ 3. Tangent to the loop
 - ☒ 4. Upward from the plane

Q.10 A planet rotates with an angular momentum of $9 \times 10^{33} \text{ kg m}^2/\text{s}$. If the moment of inertia doubles but angular momentum is conserved, what happens to its angular speed?

- Ans**
- ☒ 1. It remains unchanged
 - ☒ 2. It doubles
 - ☒ 3. It becomes half
 - ☒ 4. It becomes one-fourth

Q.11 Two wires of equal length and different materials have resistivities $\rho_1 = 2 \times 10^{-8} \Omega\text{m}$ and $\rho_2 = 6 \times 10^{-8} \Omega\text{m}$. If both have the same area, what is the ratio of their resistances R_1/R_2 ?

- Ans**
- ☒ 1. $1:3$
 - ☒ 2. $1:2$
 - ☒ 3. $3:1$
 - ☒ 4. $2:1$

Q.12 If the amplitude of a simple harmonic motion (SHM) is doubled, by what factor does the maximum kinetic energy increase?

- Ans**
- ☒ 1. 4
 - ☒ 2. 8
 - ☒ 3. 1
 - ☒ 4. 2

Q.13 Two AC circuits are connected to the same alternating voltage source. If the impedance of Circuit A is greater than that of Circuit B, how will the currents flowing through the two circuits compare?

- Ans**
- ☒ 1. The current in Circuit B is greater than in Circuit A
 - ☒ 2. The current in Circuit A is greater than in Circuit B
 - ☒ 3. The currents in both circuits are equal
 - ☒ 4. The current depends only on the frequency of the source and not on impedance

Q.14 A metal ball just passes through a ring when both are at the same temperature. If only the ball is heated while the ring remains at the same temperature, what will happen when the ball is tried again?

- Ans**
- ☐ 1. The ball passes more easily
 - ☐ 2. The ball will shrink and fit
 - ☒ 3. The ball will not pass through the ring
 - ☐ 4. The ring will expand to fit the ball

Q.15 A car crashes into a wall and comes to rest in 0.2 seconds. Which action would reduce the force experienced by the passengers?

- Ans**
- ☐ 1. Increasing the mass of the passengers
 - ☐ 2. Increasing the car's speed before collision
 - ☒ 3. Increasing the time taken to stop by using airbags
 - ☐ 4. Reducing the time to stop by using airbags

Q.16 If entropy decreases in one part of a closed system, what must happen elsewhere according to the Second Law of Thermodynamics?

- Ans**
- ☐ 1. Entropy increases in every part
 - ☒ 2. Entropy must increase elsewhere by at least as much
 - ☐ 3. Entropy stays constant everywhere
 - ☐ 4. Total entropy must decrease

Q.17 How does the magnetic field at a distance r from an infinitely long wire change if the current is doubled?

- Ans**
- ☐ 1. The field at distance r quadruples
 - ☐ 2. The field at distance r remains unchanged
 - ☐ 3. The field at distance r is halved
 - ☒ 4. The field at distance r doubles

Q.18 If the root mean square speed of oxygen molecules at 300 K is approximately 483 m/s, what is the rms speed at 1200 K?

- Ans**
- ☐ 1. 332 m/s
 - ☐ 2. 241.5 m/s
 - ☒ 3. 966 m/s
 - ☐ 4. 1932 m/s

Q.19 A device requires a current of 0.25 A to operate and has a parallel combination of two resistors of resistance $80\ \Omega$ each. According to Ohm's law, what is the minimum voltage needed to run the device?

- Ans**
- ☐ 1. 0.625 V
 - ☐ 2. 0.0625 V
 - ☒ 3. 10 V
 - ☐ 4. 160 V

Q.20 A metal has a work function of 2 eV. What is the minimum energy required (in joules) to emit an electron from its surface?

- Ans**
- ☐ 1. $1.602 \times 10^{-19}\text{ J}$
 - ☒ 2. $3.204 \times 10^{-19}\text{ J}$
 - ☐ 3. $4.806 \times 10^{-19}\text{ J}$
 - ☐ 4. $6.408 \times 10^{-19}\text{ J}$

Q.21 A person lifts a box vertically upward at constant speed through a height h . What is the work done by the gravitational force during this process?

- Ans**
- ☐ 1. Cannot be determined without mass
 - ☐ 2. Zero
 - ☐ 3. Positive and equal to mgh
 - ☒ 4. Negative and equal to $-mgh$

Q.22 Which band in a solid contains electrons that are most likely to participate in electrical conduction?

- Ans**
- ☒ 1. Conduction band
 - ☐ 2. Depletion region
 - ☐ 3. Valence band
 - ☐ 4. Forbidden gap

Q.23 Two identical coils are each connected to separate galvanometers. A magnet is moved rapidly toward the first coil and slowly toward the second coil. Based on Faraday's law of electromagnetic induction, how will the induced emf in the two coils compare?

- Ans**
- ☐ 1. The second coil produces greater emf
 - ☐ 2. No emf is produced in either coil
 - ☐ 3. Both coils produce equal emf
 - ☒ 4. The first coil produces greater emf

Q.24 A simple microscope has a focal length of 4 cm. What is its magnifying power when the final image is formed at the least distance of distinct vision (25 cm)?

- Ans**
- ☐ 1. 5
 - ☐ 2. 5.25
 - ☒ 3. 7.25
 - ☐ 4. 6.25

Q.25 What do electromagnetic waves carry from one place to another?

- Ans**
- ☐ 1. Only frequency
 - ☐ 2. Charge
 - ☐ 3. Mass
 - ☒ 4. Energy

Section : Mathematics

Q.1 Which of the following correctly represents second derivative y_2 if $x^2 + y^2 = r^2$?

- Ans**
- ☒ 1. $-r^2/y^3$
 - ☐ 2. r/y^2
 - ☐ 3. $-r/y$
 - ☐ 4. r/y^3

Q.2 Which statement about the derivative of a constant function is correct?

- Ans**
- ☒ 1. The derivative is always zero.
 - ☐ 2. The derivative is always positive.
 - ☐ 3. The derivative is always negative.
 - ☐ 4. The derivative is equal to the function itself.

Q.3 If A is 3 x 3 matrix such that $\det(A) = -2$, what is the value of $\det(\text{adj}(\text{adj } A))$?

- Ans
- ☒ 1. -32
 - ☒ 2. 16
 - ☒ 3. 4
 - ☒ 4. -8

Q.4 If $\sin^{-1}(x) + \sin^{-1}(y) = \frac{\pi}{2}$, then which of the following is true for x and y?

- Ans
- ☒ 1. $x + y = 1$
 - ☒ 2. $y = \frac{\pi}{2} + x$
 - ☒ 3. $x + y = \pi$
 - ☒ 4. $x^2 + y^2 = 1$

Q.5 What is the result of $\int_0^1 x dx$?

- Ans
- ☒ 1. 0.5
 - ☒ 2. 2
 - ☒ 3. 0.25
 - ☒ 4. 1

Q.6 What is the y-intercept of the line $4x + 2y = 8$?

- Ans
- ☒ 1. 4
 - ☒ 2. 2
 - ☒ 3. -4
 - ☒ 4. -2

Q.7 Let $\vec{x} = 2i - pj + 3k$ and $\vec{y} = qi + 4j - k$. If the vector $\vec{x} + \vec{y}$ has magnitude 4, then equation of which of the following curve is satisfied by the point (p, q)?

- Ans
- ☒ 1. circle
 - ☒ 2. ellipse
 - ☒ 3. parabola
 - ☒ 4. hyperbola

Q.8 If the arithmetic mean (AM) of 5 and x is 8, what is the value of x?

- Ans
- ☒ 1. 11
 - ☒ 2. 13
 - ☒ 3. 5
 - ☒ 4. 8

Q.9 What is the value of $\frac{\sin(x+y) - \sin(x-y)}{\cos(x+y) + \cos(x-y)}$ if $x = 45^\circ$ and $y = 30^\circ$?

- Ans
- ☒ 1. $\frac{1}{2}$
 - ☒ 2. $\frac{1}{\sqrt{3}}$
 - ☒ 3. $\frac{\sqrt{3}}{2}$
 - ☒ 4. $\frac{2}{\sqrt{3}}$

Q.10 If $n!$ denote the factorial of number n , then the value of $\frac{7!}{3!}$ is

- Ans
- ☐ 1. 420
 - ☐ 2. 1680
 - ☒ 3. 840
 - ☐ 4. 210

Q.11 Which of the following sets is empty?

- Ans
- ☐ 1. $\{x \in \mathbb{R} : x^3 < 0\}$
 - ☐ 2. $\{x \in \mathbb{R} : x^2 > 0\}$
 - ☒ 3. $\{x \in \mathbb{R} : x^2 < 0\}$
 - ☐ 4. $\{x \in \mathbb{R} : x^3 > 0\}$

Q.12 Evaluate $\int_0^1 \frac{dx}{x + \sqrt{1-x^2}}$.

- Ans
- ☐ 1. $\frac{\pi}{3}$
 - ☒ 2. $\frac{\pi}{4}$
 - ☐ 3. $\frac{\pi}{2}$
 - ☐ 4. $\frac{\pi}{6}$

Q.13 If the second derivative at a critical point is negative, what can be concluded about the point?

- Ans
- ☐ 1. Nothing can be concluded
 - ☐ 2. It is an inflection point
 - ☒ 3. It is a local maximum
 - ☐ 4. It is a local minimum

Q.14 What is the value of $\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$?

- Ans
- ☒ 1. $\ln(a)$
 - ☐ 2. a
 - ☐ 3. 1
 - ☐ 4. 0

Q.15 If $z = 4 - 5i$, what is the conjugate of z ?

- Ans
- ☒ 1. $4 + 5i$
 - ☐ 2. $-4 + 5i$
 - ☐ 3. $4 - 5i$
 - ☐ 4. $-4 - 5i$

Q.16 If a point P on the parabola $y^2 = 8x$ has focal distance 5, what is the x coordinate of P ?

- Ans
- ☐ 1. 2
 - ☒ 2. 3
 - ☐ 3. 5
 - ☐ 4. 7

Q.17 What is the value of the determinant if all elements in a row of a determinant are zero?

- Ans
- ☒ 1. One
 - ☒ 2. Zero
 - ☒ 3. Unchanged
 - ☒ 4. Negative one

Q.18 A die is rolled. Let event A be getting a number less than 4, and event B be getting a number greater than 3. Are A and B mutually exclusive, and why?

- Ans
- ☒ 1. Yes, because no outcome is both less than 4 and greater than 3
 - ☒ 2. No, because the events can occur together
 - ☒ 3. No, because 3 is in both events
 - ☒ 4. Yes, because their probabilities add up to 1

Q.19 If the circle $x^2 + y^2 + 4x + py + 1 = 0$ has its center on the line $2x - y + 1 = 0$, what is the value of p?

- Ans
- ☒ 1. 7
 - ☒ 2. 3
 - ☒ 3. 6
 - ☒ 4. 1

Q.20 If $f(x) = \sin(\cos^2(\pi x))$, find the value of $f'\left(\frac{1}{4}\right)$.

- Ans
- ☒ 1. $-\pi \cos\left(\frac{1}{2}\right)$
 - ☒ 2. $-\pi \sin\left(\frac{1}{2}\right) \cos\left(\frac{1}{2}\right)$
 - ☒ 3. $-\pi \sin\left(\frac{1}{2}\right)$
 - ☒ 4. 0

Q.21 If A is a 2x2 matrix and B is a 2x2 matrix, which statement is correct regarding AB?

- Ans
- ☒ 1. AB is a 2x2 matrix.
 - ☒ 2. AB is a 4x4 matrix.
 - ☒ 3. AB is always equal to BA.
 - ☒ 4. AB is not defined.

Q.22 Evaluate $\int \frac{1}{\sqrt{x+2} + \sqrt{x-1}} dx$.

- Ans
- ☒ 1. $\frac{1}{3} \left[(x+2)^{\frac{3}{2}} + (x-1)^{\frac{3}{2}} \right] + c$
 - ☒ 2. $\frac{2}{9} \left[(x+2)^{\frac{3}{2}} - (x-1)^{\frac{3}{2}} \right] + c$
 - ☒ 3. $\frac{1}{3} \left[(x+2)^{\frac{1}{2}} + (x-1)^{\frac{1}{2}} \right] + c$
 - ☒ 4. $\frac{2}{9} \left[(x+2)^{\frac{3}{2}} + (x-1)^{\frac{3}{2}} \right] + c$

Q.23 Which term in the expansion of $\left(x^2 + \frac{1}{x}\right)^{12}$ is independent of x ?

- Ans
- ☐ 1. 6th
 - ☒ 2. 9th
 - ☐ 3. 7th
 - ☐ 4. 8th

Q.24 A tank is draining such that the rate of change of water volume V with respect to time t is constant $\frac{dV}{dt} = -5$. What is the general solution for V ?

- Ans
- ☐ 1. $V = -t + C$
 - ☒ 2. $V = -5t + C$
 - ☐ 3. $V = 5t + C$
 - ☐ 4. $V = t + C$

Q.25 If you multiply each value in a data set by 3, how does the variance change?

- Ans
- ☐ 1. It becomes half
 - ☒ 2. It becomes 9 times the original variance
 - ☐ 3. It becomes 3 times the original variance
 - ☐ 4. It remains unchanged

Section : English

Q.1 Which word best suits the noun form of the word 'defend'?

- Ans
- ☐ 1. Defensible
 - ☐ 2. Defends
 - ☒ 3. Defence
 - ☐ 4. Defensive

Q.2 Identify the part of the sentence that contains an error.

Each of the students have reached the auditorium on time.

- Ans
- ☐ 1. Each
 - ☐ 2. reached
 - ☒ 3. have
 - ☐ 4. of the students

Q.3 Convert the following passive sentence into active voice.

The book was written by a famous author last year.

- Ans
- ☒ 1. A famous author wrote the book last year.
 - ☐ 2. A famous author had written the book last year.
 - ☐ 3. The book wrote a famous author last year.
 - ☐ 4. A famous author writes the book last year.

Q.4 Choose the correct one-word substitution for the highlighted phrase in the given sentence.

The young writer is admired for her use of **exaggerated expressions** for effect in portraying emotions.

- Ans**
- ☒ 1. satire
 - ☒ 2. hyperbole
 - ☒ 3. irony
 - ☒ 4. metaphor

Q.5 Choose the correct word to complete the sentence.

The students were instructed to remain _____ during the examination.

- Ans**
- ☒ 1. quit
 - ☒ 2. quite
 - ☒ 3. quilt
 - ☒ 4. quiet

Q.6 Select the most appropriate option which is the indirect speech of the given sentence.

Manoj said to Shyam, "What are you doing here?"

- Ans**
- ☒ 1. Manoj asked Shyam what was he doing there.
 - ☒ 2. Manoj asked Shyam what he is doing here.
 - ☒ 3. Manoj said to Shyam what he was doing here.
 - ☒ 4. Manoj asked Shyam what he was doing there.

Q.7 Rectify the sentence by selecting the correct spelling of the underlined word.

The kids were excited to explore the new collection in the libary.

- Ans**
- ☒ 1. librery
 - ☒ 2. lybrary
 - ☒ 3. library
 - ☒ 4. laibrary

Q.8 The following sentence has been divided into parts. One of them may contain an error. Select the part that contains the error from the given options.

I suggested that she goes to the doctor immediately, since she was feeling unwell.

- Ans**
- ☒ 1. I suggested
 - ☒ 2. that she goes
 - ☒ 3. to the doctor immediately
 - ☒ 4. since she was feeling unwell

Q.9 Choose the correct option to fill in the blank.

All I need is _____ hour and I'll be finished with _____ work.

- Ans**
- ☒ 1. an, the
 - ☒ 2. the, a
 - ☒ 3. a, an
 - ☒ 4. an, an

Q.10 The first part of the sentence is in the correct position. The other parts (P, Q, R, S) are mixed up. Put them in the right order to make a meaningful sentence.

The rain started
P: cooling the hot
Q: weather in the afternoon
R: falling as dark clouds
S: gathered over the city

- Ans
- ☒ 1. SQPR
 - ☒ 2. RSPQ
 - ☒ 3. PSRQ
 - ☒ 4. QRSP

Q.11 Fill in the blank with the correct option.

_____ is reading a book quietly.

- Ans
- ☒ 1. Him
 - ☒ 2. Her
 - ☒ 3. Me
 - ☒ 4. She

Q.12 Select the option that is the opposite of the underlined word in the given sentence.

The judge delivered a harsh verdict after the trial.

- Ans
- ☒ 1. Concise
 - ☒ 2. Severe
 - ☒ 3. Ruthless
 - ☒ 4. Lenient

Q.13 Identify the error in the following sentence and choose the option that corrects it:

The researchers analyse the data carefully before they submitted the final report.

- Ans
- ☒ 1. analysed
 - ☒ 2. analyses
 - ☒ 3. has analyse
 - ☒ 4. analysing

Q.14 Rectify the given sentence by selecting the correct option for the underlined word.

Lisa's farm is just through the bridge.

- Ans
- ☒ 1. from
 - ☒ 2. above
 - ☒ 3. to
 - ☒ 4. beyond

Q.15 Identify the segment of the sentence that contains a grammatical error.

By the time the conference starts (P) / she will be finishing (Q) / her preparations (R) / perfectly. (S)

- Ans
- ☒ 1. P
 - ☒ 2. Q
 - ☒ 3. S
 - ☒ 4. R

Q.16 Select the option that has the closest meaning to the underlined word in the sentence.

My colleague always spoke in a polite manner to her students.

- Ans**
- ☒ 1. Angry
 - ☒ 2. Rude
 - ☒ 3. Careless
 - ☒ 4. Gentle

Comprehension:

Read the given passage and answer the questions that follow.

Microplastics were once thought to be only an ocean problem. But now, scientists have found them in the air, soil, and even drinking water. Studies show that these tiny plastic pieces can travel long distances through wind and water, making it almost impossible to avoid them. Because of this, scientists are now studying how microplastics affect humans, animals, and the environment—not just oceans. Researchers are trying to understand how these particles behave inside living organisms. Early research shows that most microplastics pass out of the body, but some may stay in tissues or carry harmful chemicals. However, scientists still don't fully know the long-term effects because many studies are small. Microplastics are now seen as signs of deeper problems in how we produce, use, and throw away plastic. Scientists warn people not to panic—finding microplastics does not automatically mean they are toxic. But their presence everywhere raises questions about waste, pollution, and the durability of plastic. The issue is no longer a local problem; it has become a global challenge, and experts say we need better materials and more responsible consumption.

SubQuestion No : 17

Q.17 Where have microplastics been found according to recent studies?

- Ans**
- ☒ 1. In air, soil, and drinking water
 - ☒ 2. Only in plastic bottles
 - ☒ 3. Only in oceans
 - ☒ 4. Only in factories

Comprehension:

Read the given passage and answer the questions that follow.

Microplastics were once thought to be only an ocean problem. But now, scientists have found them in the air, soil, and even drinking water. Studies show that these tiny plastic pieces can travel long distances through wind and water, making it almost impossible to avoid them. Because of this, scientists are now studying how microplastics affect humans, animals, and the environment—not just oceans. Researchers are trying to understand how these particles behave inside living organisms. Early research shows that most microplastics pass out of the body, but some may stay in tissues or carry harmful chemicals. However, scientists still don't fully know the long-term effects because many studies are small. Microplastics are now seen as signs of deeper problems in how we produce, use, and throw away plastic. Scientists warn people not to panic—finding microplastics does not automatically mean they are toxic. But their presence everywhere raises questions about waste, pollution, and the durability of plastic. The issue is no longer a local problem; it has become a global challenge, and experts say we need better materials and more responsible consumption.

SubQuestion No : 18

Q.18 Why do scientists say microplastics are now a global issue?

- Ans**
- ☒ 1. They are easy to clean up
 - ☒ 2. They travel long distances and appear everywhere
 - ☒ 3. They disappear quickly in nature
 - ☒ 4. Only animals are affected

Comprehension:

Read the given passage and answer the questions that follow.

Microplastics were once thought to be only an ocean problem. But now, scientists have found them in the air, soil, and even drinking water. Studies show that these tiny plastic pieces can travel long distances through wind and water, making it almost impossible to avoid them. Because of this, scientists are now studying how microplastics affect humans, animals, and the environment—not just oceans. Researchers are trying to understand how these particles behave inside living organisms. Early research shows that most microplastics pass out of the body, but some may stay in tissues or carry harmful chemicals. However, scientists still don't fully know the long-term effects because many studies are small. Microplastics are now seen as signs of deeper problems in how we produce, use, and throw away plastic. Scientists warn people not to panic—finding microplastics does not automatically mean they are toxic. But their presence everywhere raises questions about waste, pollution, and the durability of plastic. The issue is no longer a local problem; it has become a global challenge, and experts say we need better materials and more responsible consumption.

SubQuestion No : 19

Q.19 The word “contaminants” in the passage means:

- Ans**
- ☒ 1. Things that produce energy
 - ☒ 2. Things that clean the environment
 - ☒ 3. Things that improve water quality
 - ☒ 4. Things that make something dirty or unsafe

Comprehension:

Read the given passage and answer the questions that follow.

Microplastics were once thought to be only an ocean problem. But now, scientists have found them in the air, soil, and even drinking water. Studies show that these tiny plastic pieces can travel long distances through wind and water, making it almost impossible to avoid them. Because of this, scientists are now studying how microplastics affect humans, animals, and the environment—not just oceans. Researchers are trying to understand how these particles behave inside living organisms. Early research shows that most microplastics pass out of the body, but some may stay in tissues or carry harmful chemicals. However, scientists still don't fully know the long-term effects because many studies are small. Microplastics are now seen as signs of deeper problems in how we produce, use, and throw away plastic. Scientists warn people not to panic—finding microplastics does not automatically mean they are toxic. But their presence everywhere raises questions about waste, pollution, and the durability of plastic. The issue is no longer a local problem; it has become a global challenge, and experts say we need better materials and more responsible consumption.

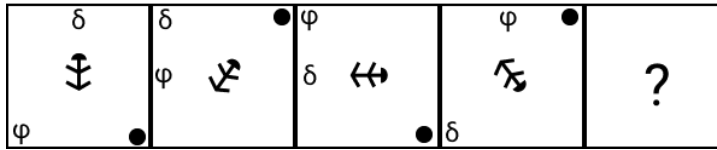
SubQuestion No : 20

Q.20 Which of the following statements is TRUE according to the passage?

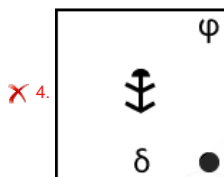
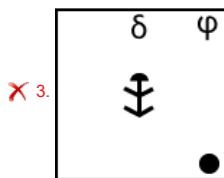
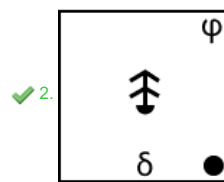
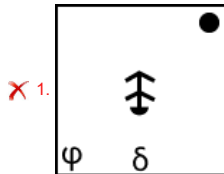
- Ans**
- ☒ 1. Most microplastics pass through the body, but some may stay in tissues.
 - ☒ 2. Microplastics can be avoided easily with simple steps.
 - ☒ 3. Detecting microplastics means they are always toxic.
 - ☒ 4. Scientists have already fully understood all long-term effects.

Section : RAGA

Q.1 Identify the figure given in the options which when put in place of '?' will logically complete the series?



Ans



Q.2 A triangle has an angle equal to 90° and two sides of equal lengths. If the length of unequal side is $18\sqrt{2}$ cm, then the area (in cm^2) is:

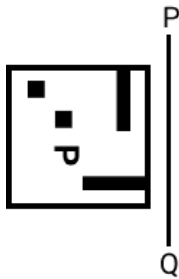
Ans ✓ 1. 162

✗ 2. 166

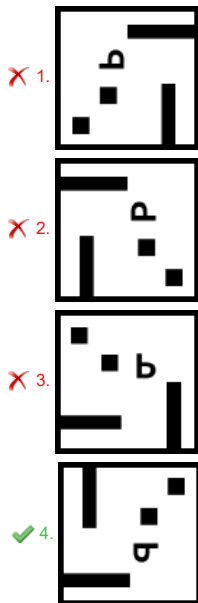
✗ 3. 160

✗ 4. 164

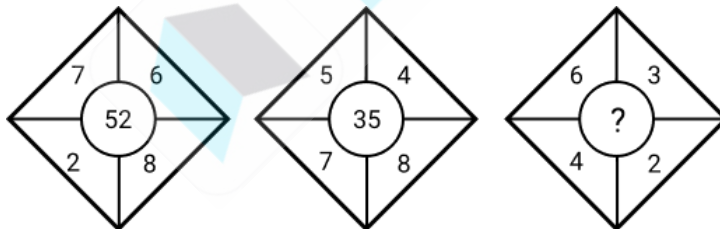
Q.3 Select the correct mirror image of the given combination when the mirror is placed at 'PQ' as shown below.



Ans



Q.4 Study the given pattern carefully and select the number that can replace the question mark (?) in it.



Ans

✗ 1. 21

✗ 2. 27

✓ 3. 24

✗ 4. 32

Q.5 Seven people - H, K, P, R, A, L and V - are sitting around a circular table, facing the centre. R is the immediate neighbour of V and P. K sits second to the right of P. H is the immediate neighbour of A and V. What is the position of L with respect to A?

Ans

✗ 1. Second to the right

✓ 2. Second to the left

✗ 3. Third to the right

✗ 4. Immediate right

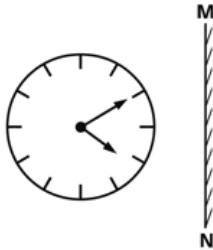
Q.6 Six people — M, T, B, U, L and E - scored different marks in post-graduation. E did not score more than B. M was the third highest scorer. L was the second lowest scorer. U scored more than B but less than T. How many people scored less than E?

- Ans**
- ☒ 1. None
 - ☐ 2. Five
 - ☐ 3. Four
 - ☐ 4. Two

Q.7 Who was the Portuguese navigator who landed at Kappad near Calicut in 1498, initiating direct maritime trade between Europe and India?

- Ans**
- ☐ 1. Dupleix
 - ☐ 2. Alfonso de Albuquerque
 - ☒ 3. Vasco da Gama
 - ☐ 4. Francois Martin

Q.8 What will be the time shown in the mirror image of the given clock when the mirror is placed along MN?



- Ans**
- ☐ 1. 09:50 hrs
 - ☒ 2. 07:50 hrs
 - ☐ 3. 02:50 hrs
 - ☐ 4. 08:50 hrs

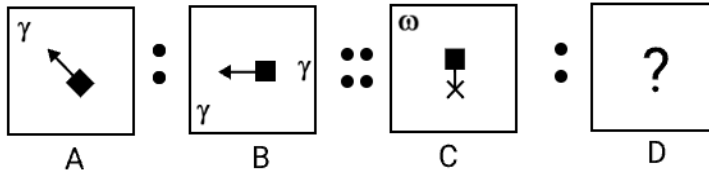
Q.9 Which gadget has data input and output capabilities?

- Ans**
- ☐ 1. Keyboard
 - ☐ 2. Scanner
 - ☐ 3. Monitor
 - ☒ 4. Touchscreen

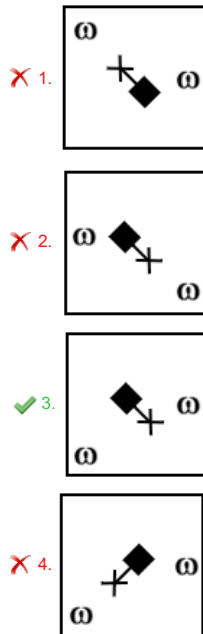
Q.10 Where was the 2025 G20 Summit held?

- Ans**
- ☐ 1. Riyadh
 - ☐ 2. Tokyo
 - ☐ 3. Berlin
 - ☒ 4. Johannesburg

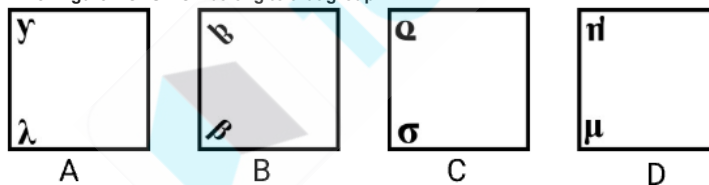
Q.11 Figure A is related to B in a certain pattern. Following the same pattern, figure C is related to D. Study the pattern and select the figure which should be placed in place of D.



Ans



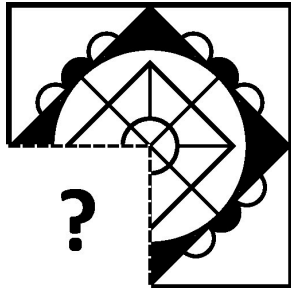
Q.12 Three of the following four figures are alike in a certain way and thus form a group. Which figure DOES NOT belong to that group?



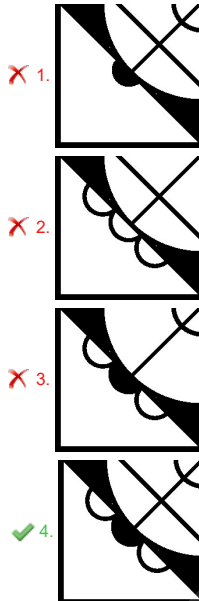
Ans

- ✗ 1. A
- ✗ 2. B
- ✗ 3. C
- ✓ 4. D

Q.13 Select the option figure that will replace the question mark (?) in the figure given below to complete the pattern.



Ans



Q.14

$$\text{If } X - 2.5 = 1 + \frac{8.7 - \left[7.6 - \left\{ \frac{13}{2} - (5.4 - 23 \div 10) \right\} \right]}{\left[1.25 \div \frac{3}{2} \div \left(\frac{1}{15} + 1 - 0.9 \right) \right] \div (2.5 \times 4)}, \text{ then value of X is:}$$

- Ans
- ✓ 1. 12.5
 - ✗ 2. 13.5
 - ✗ 3. 11
 - ✗ 4. 14

Q.15 Which plant tissue primarily functions in the storage of food?

- Ans
- ✗ 1. Stomata
 - ✗ 2. Trichomes
 - ✗ 3. Epithelial Tissue
 - ✓ 4. Parenchyma

Q.16 The average price of three toy items is ₹4500. If their prices are in the ratio 3:5:7, the price of the cheapest item (in ₹) is:

- Ans
- ✓ 1. 2700
 - ✗ 2. 900
 - ✗ 3. 1800
 - ✗ 4. 6300

Q.17 Who became the first woman to be appointed as Member (Human Resources) on the Board of the Airports Authority of India (AAI) in January 2026?

- Ans
- ✗ 1. Shikha Rani
 - ✗ 2. Radha Bhatia
 - ✗ 3. Usha Padhee
 - ✓ 4. Nivedita Dubey

Q.18 In a certain code language, "TIME" is coded as 7-18-14-22 and "LOVE" is coded as 15-12-5-22. What is the code for "CARE" in that code language?

- Ans
- ✓ 1. 24-26-9-22
 - ✗ 2. 24-26-15-22
 - ✗ 3. 24-26-18-22
 - ✗ 4. 7-26-9-22

Q.19 In a mathematical code, if '\$' means '+', '#' means '-', '@' means 'x', and '%' means '÷', find the value of Z in the following equation.

$$20 \% 5 @ 4 \$ Z = 20$$

- Ans
- ✗ 1. 8
 - ✗ 2. 2
 - ✓ 3. 4
 - ✗ 4. 6

Q.20 What should come in place of the '?' in the given series based on the English alphabetical order?

Q S X N P U K M R H J O ?

- Ans
- ✓ 1. EGL
 - ✗ 2. ELG
 - ✗ 3. EGF
 - ✗ 4. EGY

Q.21 Based on the English alphabetical order, three of the following four are alike in a certain way and thus form a group. Which is the one that does not belong to that group? (Note: The odd one out is not based on the number of consonants/vowels or their position in the letter cluster.)

- Ans
- ✗ 1. YUR
 - ✗ 2. SOL
 - ✓ 3. LHD
 - ✗ 4. DZW

Q.22 Find the lowest six-digit number that is exactly divisible by 8, 10, 12 and 16.

- Ans
- ✗ 1. 100020
 - ✗ 2. 100040
 - ✓ 3. 100080
 - ✗ 4. 100060

Q.23 Which ideal mentioned in the Preamble of India's Constitution assures brotherhood and unity among citizens?

- Ans
- ☒ 1. Sovereignty
 - ☒ 2. Fraternity
 - ☒ 3. Justice
 - ☒ 4. Liberty

Q.24 A boy starts by walking 30 m towards the West. He then turns 45° to his right and walks 10 m. From there, he turns 135° to his left and walks 18 m. He again turns 90° to his right and walks 12 m. In which direction is he finally facing?

- Ans
- ☒ 1. West
 - ☒ 2. North-East
 - ☒ 3. South
 - ☒ 4. East

Q.25 If the third proportional of 15 and 60 be x, then what is the value of x?

- Ans
- ☒ 1. 240
 - ☒ 2. 242
 - ☒ 3. 239
 - ☒ 4. 243

Q.26 The diameter of the base of a right circular cone is 28 cm and its height is $\sqrt{29}$ cm. What is the curved surface area (in cm^2) of the right circular cone? (Use $\pi = \frac{22}{7}$)

- Ans
- ☒ 1. 660
 - ☒ 2. 625
 - ☒ 3. 624
 - ☒ 4. 676

Q.27 If $12\sin A + \cos A = \sqrt{13} \sin A$, find the value of $\tan A$.

- Ans
- ☒ 1. $\frac{-12 - \sqrt{13}}{136}$
 - ☒ 2. $\frac{-13 - \sqrt{13}}{131}$
 - ☒ 3. $\frac{-12 - \sqrt{13}}{141}$
 - ☒ 4. $\frac{-12 - \sqrt{13}}{131}$

Q.28 DEAL is related to IHFO in a certain way based on the English alphabetical order. In the same way, PLAY is related to UOFB. To which of the following options is LAST related, following the same logic?

- Ans
- ☒ 1. DSWE
 - ☒ 2. QDXW
 - ☒ 3. QSXD
 - ☒ 4. QASW



Q.29 Who is the author of the short story collection, Heart Lamp, which won the International Booker Prize 2025?

- Ans
- ☐ 1. Namita Gokhale
 - ☐ 2. Geetanjali Shree
 - ☐ 3. Amitav Ghosh
 - ☒ 4. Banu Mushtaq

Q.30 Which of the following rivers lies at the eastern end of the Himalayas before the mountains bend south beyond the Dihang Gorge?

- Ans
- ☐ 1. Narmada River
 - ☐ 2. Indus River
 - ☒ 3. Brahmaputra River
 - ☐ 4. Godavari River





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